Embedded Systems Development

Exercise 5

1. Design a "reaction timer" using ATmega328P microcontroller and give your design as a schematic diagram.

   The "reaction timer" is an embedded system that measures the user's reaction time to a visual signal and displays it in a scale of three LEDs. When the user presses the "ready" pushbutton, the "reaction timer" turns off all LEDs and waits for a random amount of time between 1 and 10 seconds. Upon the expiration of this waiting time, the "reaction timer" turns on a blue LED to visually signal the user. The user must then press the "react" pushbutton as quickly as possible to react to the signal. The "reaction timer" then turns off the blue LED used to signal the user and measures the elapsed time between the visual signal and the user's reaction to the signal. If the user's reaction time to the signal is less than one second, one second to less than two seconds or two seconds or greater, the "reaction timer" turns on a green, orange or a red LED, respectively to display the user's reaction performance.

   Schematic Diagram

2. If the microcontroller is clocked at 16MHz and the timer prescale is 1024, what is the initial value required to configure a 16-bit timer to overflow in one second?

3. Write the required firmware for the "reaction timer" in C.
   Hint: Use *srand()* function defined in *stdlib.h* and an 8-bit timer to seed the random number generator with a fairly random seed, *rand()* function defined in the same header file to generate random numbers and a 16-bit timer with a numeric variable to measure the waiting time to the signal as well as the user's reaction time.

4. Modify the firmware you provided for (3) above using a floating analog input as the source for a fairly random seed to seed the random number generator.

5. Rewrite the firmware you provided for (4) above using the Arduino wrapper functions.

6. Compare the two firmware implementations that you provided for (4) and (5) above.